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DeepSeek’s Chinese-style Innovation: A New Form of AI Economics (Part 1)

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DIGITALIZATION AND TECHNOLOGY

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This insightful four-part series by CKGSB Professor [Chen Long](#) examines how [DeepSeek](#), a trailblazer in Chinese-style AI, is redefining the economics of AI and scaling laws while setting new global benchmarks. From its unprecedented surge in daily active users to groundbreaking technological advancements and strategic execution, DeepSeek presents a compelling case study on the future of AI.

The series includes four parts:

- PART 1 | The arrival of a new global AI model with the highest daily active users
- PART 2 | What has DeepSeek achieved?
- PART 3 | How did DeepSeek accomplish this?
- PART 4 | What does the “DeepSeek Phenomenon” imply?

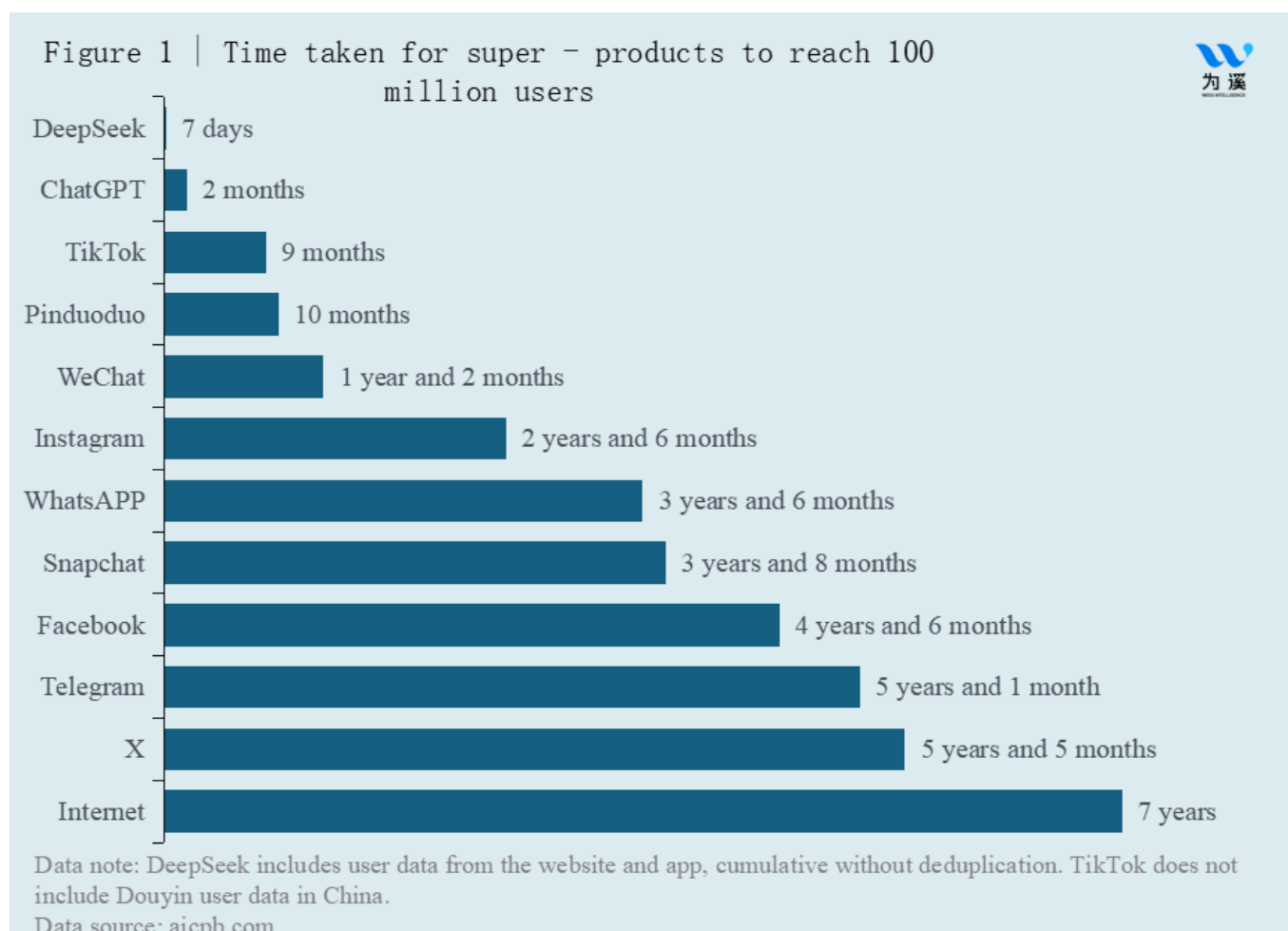
Stay tuned as we uncover how DeepSeek’s innovations are redefining the AI industry and shaping its global trajectory.

PART 1 | The Arrival of a New Global AI Model with Highest Daily Active Users

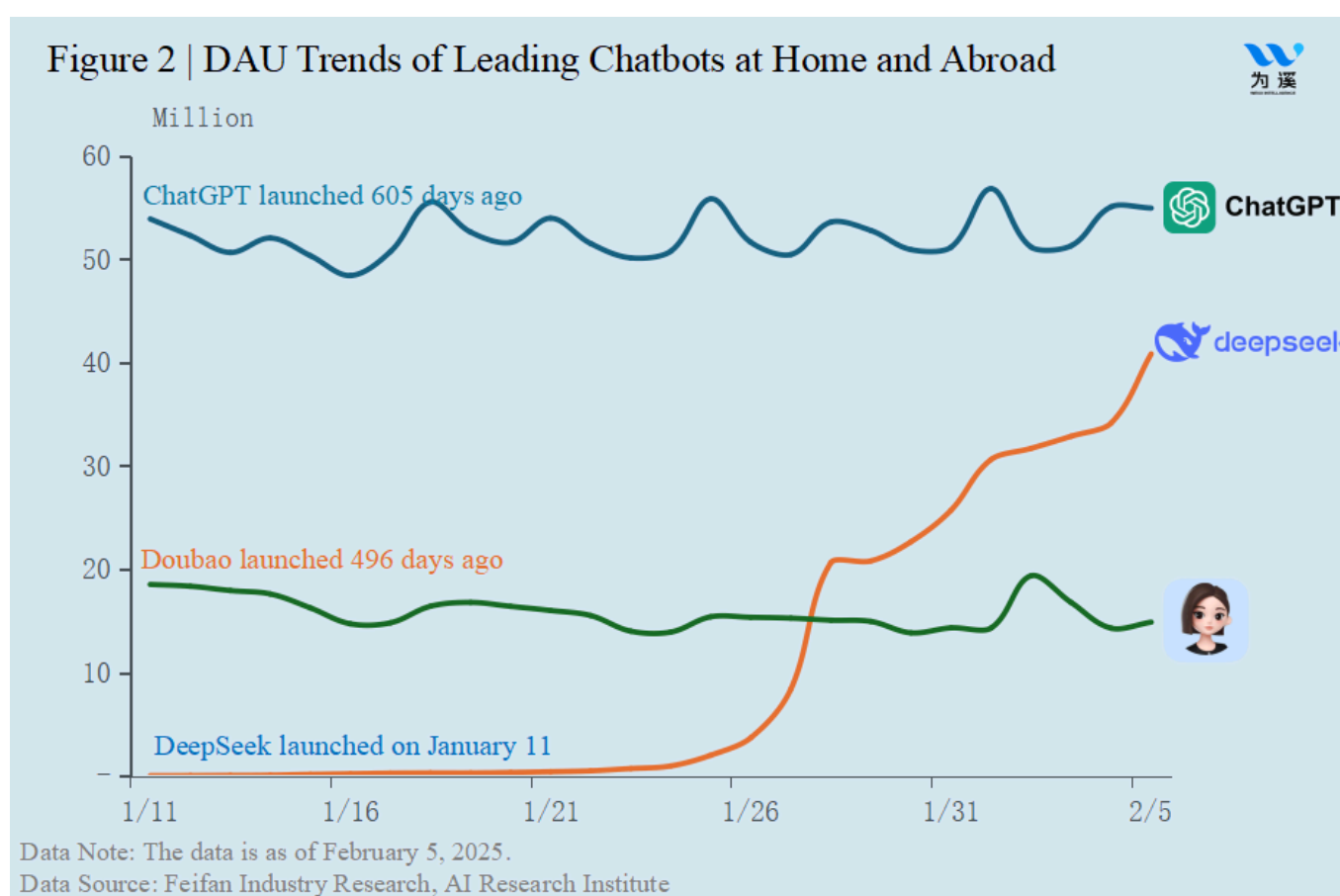
01 | The “DeepSeek Phenomenon” in the Age of AI Revolution

On January 24, 2025, the Chinese Artificial Intelligence (AI) app DeepSeek-R1 was launched. Only two weeks later, it had already overtaken ChatGPT and other rivals to become the top-rated free application on Apple’s App Store.

The explosive growth of its user base and daily active users has validated this miracle. Without any advertising efforts, by the end of January, DeepSeek had accumulated 125 million users, with over 80% of them coming in the final week, marking a growth of 100 million users in just 7 days. This made DeepSeek the fastest-growing app in history in terms of user accumulation. By comparison, the previous record-holder, ChatGPT, took two months to reach 100 million users (see Figure 1).



As of January 31, 2025, DeepSeek's daily active users (DAU) surpassed 22.15 million. In just over a week, it exceeded the efforts of Doubao, which had 16.95 million DAU in a year, reaching 40% of ChatGPT's DAU (53.23 million, as shown in Figure 2). By February 5, DeepSeek's DAU had exceeded 40.83 million, 2.75 times that of Doubao and approximately 75% of ChatGPT's DAU. Projections indicate that DeepSeek will soon surpass ChatGPT, becoming the AI model with the highest daily active users in the world.



In just two weeks since its launch, DeepSeek has already emerged as a product of the same scale as ChatGPT, and following its rival, DeepSeek has become the second phenomenon-level product in this round of the AI revolution.

On January 27, 2025, DeepSeek topped the US App download charts. On the same day, Nvidia, a global AI leader, saw its stock price plunge by 17%, wiping out nearly \$600 billion in market value, and setting the record for the largest single-day market value loss in U.S. stock market history. Meanwhile, TSMC's stock also dropped by 13%, losing more than \$150 billion in market value. Why did this series of market fluctuations occur?

These market reactions reflect the impact of DeepSeek's rise on the global AI landscape. AI technology is shaping the world through phenomenon-level breakthroughs. How these new phenomenon-level products affect business and the world becomes critically important.

As the first phenomenon-level product, ChatGPT emerged in December 2022 and led the U.S.-dominated AI revolution over the next two years. This revolution manifested in the prosperity of a new AI industrial landscape, represented by the "AI Seven Sisters"—the seven major U.S. tech giants—which span chipmakers, large models, and industrial applications. The AI Seven Sisters include Microsoft, Nvidia, Apple, Google, Amazon, Meta (Facebook), and Tesla. These companies control the most advanced computational power leaders, cutting-edge large models, and the most mature digital ecosystems with the richest data applications. Additionally,

because of their massive user bases, they play a crucial role in popularizing new AI technologies. We can summarize that the most significant feature of the previous stage of the intelligence revolution was the “ChatGPT phenomenon”, i.e., the global AI industry landscape led by ChatGPT and dominated by the United States.

It is precisely because of the U.S.’s global leadership in the existing AI industry that the U.S. government introduced the “*U.S. Framework for Artificial Intelligence Diffusion*” in January 2025. The aim of the framework is to regulate the diffusion of AI among allied nations, controllable states, and restricted regions by limiting chip advancement and large model capabilities.

Nvidia’s dramatic decline (though it later recovered most of the losses) underscored the market’s anxiety over new uncertainties: Is DeepSeek’s emergence the dawn of a new phenomenon? How much will it disrupt the AI industry landscape set in motion by OpenAI?

The world has already begun to narrate this. Deutsche Bank analyst Peter Milliken said, “The emergence of DeepSeek marks China’s ‘Sputnik moment’, successfully shattering the West’s fantasy of containing China.” The so-called Sputnik moment refers to October 4, 1957, when the Soviet Union successfully launched the Sputnik 1 artificial satellite ahead of the United States, causing a period of fear and anxiety in the Western world. Trump also pointed out, “DeepSeek has sent a clear warning to related industries in the U.S.; America needs to focus on winning this competition.”

02 | The DeepSeek Phenomenon and the Imminent Future

As the DeepSeek phenomenon continues to unfold, what future does this new wave signal?

In what seems to be an answer to this question, in the early hours of February 10, 2025, OpenAI CEO Sam Altman penned an article outlining the three AI laws that, in his view, have been validated and will determine the past and future of AI:

- **First, the intelligence of AI models is logarithmically proportional to the resources invested.** The intelligence of an AI model is roughly equal to the logarithm of the resources used to train and run it. The amount you invest can roughly predict the return you will get. This prediction is accurate across several orders of magnitude.
- **Second, the rapid decline in the cost of using AI and the exponential growth in demand.** The cost of using AI decreases approximately 10-fold every 12 months, and lower prices lead to more demand. From the beginning of 2023 with GPT-4 to mid-2024 with GPT-4o, the price per token dropped by about 150 times, clearly evidencing the scaling law. While Moore’s Law used to change the world at a doubling rate every 18 months; now, the scaling laws shown by AI are incredibly more powerful.
- **Third, the linear growth of AI leads to exponential growth in socioeconomic value.** This growth trend indicates that investment in AI will continue to grow at a high rate in the future, and there is no reason to believe that exponential investment growth will stop in the near term.

Altman is saying that these three laws together define how AI technology will change the economy and business: First, in terms of technical capabilities, the investment in AI technology determines its capabilities. Second, in terms of application iteration speed, AI capabilities and demand create a positive feedback loop, reflected in rapidly decreasing costs and higher demand, which can be measured by the “scaling law” (Scaling Law, the scaling law of the past half-century being Moore’s Law). Lastly, because AI is a general-purpose technology, the economic and business value it brings will grow exponentially.

Taken together, **Technology → Scaling Law → Value Creation** is Altman’s view of AI economics, or the laws that define how AI technology impacts the economy and business.

Altman’s description reveals the key patterns of AI technology development, but what is notably missing from his description (which current leaders may either deliberately or inadvertently overlook) is that the scale of technological investment and the affordability of technological products actually determine the number of industry participants, who will dominate these three laws, and the resulting industry structure. OpenAI’s development path, which stresses how “significant efforts lead to miracles,” has formed an oligopolistic business landscape centered around major U.S. tech companies. The three laws Altman speaks of are carried and led by these companies, with clear regional and oligopolistic characteristics.

However, the emergence of DeepSeek is changing this status quo, and this is the significance of the DeepSeek phenomenon. Much of the investment cost is actually the cost of exploring paths. **New Chinese AI players, represented by DeepSeek and Qwen, are offering clearer technological paths, significantly reducing both development and application costs, and embracing more open-source and shared models to reshape the AI industry. These factors will fundamentally change the logic of the previous stage of AI economics, thus altering the way and pace of AI diffusion. The laws that were once based on oligopolistic core players are evolving into Inclusive Scaling Law, which will lead to a very different global AI industry landscape.** Pandora’s box has been opened. The world is about to enter a “DeepSeek phenomenon” phase.

We will use two articles to explain the logic and trends behind this. In this series, we will first explain what DeepSeek has achieved, examining how it has accomplished this, and how it changes AI economics, including the scaling law. In the second article, we will attempt to analyze how the DeepSeek phenomenon will affect various sectors- including chips, cloud computing, large models, and AI industry applications, – while exploring some of the key trends that are beginning to emerge.

Through a detailed analysis of the DeepSeek phenomenon in this series, we believe we are witnessing the unfolding of the future of AI – an evolution driven by DeepSeek’s disruptive force.